

Enhancing Student Engagement Through AI-Powered Educational Chatbots: A Retrieval-Augmented Generation Approach

Artificial Intelligence (AI) has rapidly advanced, significantly impacting education through personalized learning solutions. This paper presents the development and evaluation of a novel Retrieval-Augmented Generation (RAG) system designed to deliver tailored educational content based on individual learner progress and needs. Leveraging open-source large language models (LLMs) such as Gemma2, Mistral, and Llama3.2, the system ensures transparency, flexibility, and robust data privacy by enabling local hosting within educational institutions. The architecture comprises a Vue.js and Tailwind CSS-based frontend, a Node.js and Ollama Framework-integrated backend, and a PostgreSQL database. Evaluation was conducted in two phases: validating GPT-4o as a reliable automated evaluation tool against human assessments, and assessing the performance of the selected LLMs in generating high-quality learning materials. Results indicate Gemma2 outperforms other models in accuracy, completeness, correctness, and comprehension, while GPT-4o reliably automates content evaluation. This study advances digital educational platforms by integrating state-of-the-art AI technologies with pedagogical principles, enhancing personalized learning experiences.

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